ARITRI HALDER

Minneapolis, MN | 952-220-8718 | aritrihalder2001@gmail.com | linkedin.com/in/aritri-halder

PROFESSIONAL SUMMARY

Mechanical Engineering Ph.D. candidate with hands-on experience developing mechanical and thermal-fluid systems from concept through fabrication and experimental validation. Proficient in SolidWorks, ASME Y14.5 GD&T, and tolerance stack-up analysis to develop fabrication-ready engineering drawings and assemblies. Experienced in structural and thermal FEA to evaluate component integrity and guide design improvements. Proven track record in design of experiments (DOE), root-cause analysis, and cross-functional technology transfer with industry partners, suppliers, and research laboratories. Open to relocation.

EDUCATION

Doctor of Philosophy in Mechanical Engineering Aug. 2023 - Expected Jan. 2027
University of Minnesota Twin Cities, Minneapolis, MN

- Advisor: Prof. William F. Northrop; GPA: 3.67/4.0
- *Thesis: Experimental investigation of Green Ammonia Burners for Industrial Applications.*

Master of Science in Mechanical Engineering Aug. 2023 - Sept. 2026
University of Minnesota Twin Cities, Minneapolis, MN

- *Specialization: Heat Transfer & Multiphase Flow and Reacting Systems*

Bachelor of Engineering in Mechanical Engineering Aug. 2019 - July 2023
Jadavpur University, Kolkata, India; GPA: 9.12/10.0

TECHNICAL SKILLS

- **Mechanical Design & Manufacturing:** SolidWorks, Creo, ASME Y14.5 GD&T, Tolerance stack-up analysis, AutoCAD, DFMEA, Gas Delivery Systems, Machining, Prototyping, Control systems, DOE, RCA, Automation, Data Analysis, Optical and Laser diagnostics (multi-species chemiluminescence, PIV, computed tomography), Spectroscopy, Cleanroom handling.
- **Modeling & Simulation:** SolidWorks Simulation (structural/thermal FEA), ANSYS Fluent, MATLAB, Python, C/C++, LabVIEW, image processing, MS Office, CHEMKIN and Tecplot.

RELEVANT EXPERIENCE

University of Minnesota Twin Cities, Minneapolis, USA Aug. 2023 - Present

Graduate Research Assistant - Dr. William Northrop

- Developed a novel 20 kW high-temperature system using SolidWorks to study the thermal-fluid behavior and system performance under varying operating conditions achieving 60% reduced NOx emissions.
- Released fabrication-ready SolidWorks assemblies, ASME Y14.5 engineering drawings by applying tolerance stack-up analysis and thermal/static FEA to verify assembly fit and achieve $\sim$1% stress-to-target.
- Led the 6x scale-up of the mechanical system (20-120 kW), applying systematic Design of Experiments (DOE) to characterize operating stability under controlled gas-flow rates and mixture ratios.
- Reduced data-processing time by more than 99% through automated MATLAB/Python workflows while integrating gas-flow control and LabVIEW data acquisition for repeatable thermal-fluid testing.
- Reduced costs by 60% by developing a mechanically indexed camera assembly with 20 repeatable viewing angles to characterize 3D images of flame structures, reducing equipment procurement costs by over $4500.
- Spearheaded three R&D projects with industry partners and researchers, delivering quarterly technical milestones and resulting in a top-tier journal publication and the Best Thermal Applications distinction at an international conference (300+ attendees).

Indian Institute of Technology (IIT) Kanpur, Kanpur, India May 2022 - July 2022

Research Intern

- Improved heat-exchanger performance by 42% through ANSYS Fluent optimization of minichannel geometry, enhancing heat transfer while managing pressure losses.
- Reduced temperature by 18% by evaluating minichannel configurations using finite-volume CFD with SIMPLE pressure-velocity coupling simulating cooling architecture of chips .
- Established mesh-independent heat-transfer and pressure-drop predictions across Reynolds numbers 200-900 through systematic grid-convergence studies to verify numerical solution reliability.

ADDITIONAL PROJECTS

Research Assistant, Jadavpur University Aug. 2022 - June 2023

- Ran parallel C/C++ immersed-boundary simulations on HPC at $Re = 250$ to resolve fluid-structure interaction and systematically evaluate unsteady wake response.
- Mapped wake transitions and flow separation by varying Strouhal number and pitching phase and analyzing how structural kinematics altered vortex shedding using Tecplot.

Research Assistant, Indian Institute of Technology (IIT) Madras May 2022 - Sept. 2022

- Modeled bio-inspired sharkskin surfaces with an immersed-boundary finite-difference solver to quantify effects on skin friction and boundary-layer transition for complex surface geometry.
- Identified dominant unsteady-flow frequencies using power spectral density, scalograms, and Strouhal-number scaling to connect velocity fluctuations with surface-induced flow behavior.

TEACHING & MENTORING EXPERIENCE

Teaching Assistant, Department of Mechanical Engineering University of Minnesota Twin Cities 2024, 2025, 2026

- Mentored 50+ senior undergraduate and 4+ graduate students in experimental testing, instrumentation, data acquisition, and engineering analysis involving pumps, hydrogen fuel cells, gas turbines, and heat-transfer systems.
- Supported ME 4431 Energy Conversion Laboratory and ME 3333 Heat Transfer through laboratory supervision, safety guidance, experimental troubleshooting, and instruction in conduction, convection, radiation, and thermal-resistance analysis.

AWARDS & FELLOWSHIPS

- Travel Award, 41st International Symposium on Combustion, 2026.
- Best Abstract - Thermal Applications, 4th Symposium on Ammonia Energy, 2025.
- Chester E. Dekko Foundation Endowed Graduate Fellowship, University of Minnesota, 2024-2025.
- Graduate Student Fellowship, Department of Mechanical Engineering, University of Minnesota, 2023.
- First Class Distinction, Jadavpur University, 2023.

PROFESSIONAL SERVICE & LEADERSHIP

- **Peer Reviewer, Combustion Science and Technology**, 2026: Evaluated technical manuscripts for scientific rigor and research quality.
- **Organizer, 4th Symposium on Ammonia Energy**, Minneapolis, MN, 2025: Coordinated event activities and supported engagement among academic and industry participants.
- **Social Chair, Mechanical Engineering Graduate Student Council**, University of Minnesota, 2024-2026: Organized departmental events supporting graduate-student and faculty engagement.

PUBLICATIONS

Peer-Reviewed Publications

- **A. Halder**, C. Goertemiller, A. Hoxie, and W. F. Northrop, "Experimental investigation of a 20 kW atmospheric pressure pure ammonia jet-swirl burner," *Proceedings of the Combustion Institute*, vol. 42, Art. no. 106019, 2026. doi:10.1016/j.proci.2026.106019.
- **A. Halder**, R. Pandey, and U. Madanan, "Computer simulations on heat transfer enhancement in a minichannel heat sink using modified butterfly inserts," *Proceedings of the 27th National and 5th International ISHMT-ASTFE Heat and Mass Transfer Conference*, pp. 149–154, 2024. doi:10.1615/IHMTC-2023.250.

Manuscripts in Preparation

- **A. Halder**, C. Goertemiller, and W. F. Northrop, "Stability and Emissions Profile of a 120 kW Fuel-Flexible Jet-Swirl Burner Operating on Ammonia." Manuscript in preparation for *Transformative Energy*.
- **A. Halder**, C. Goertemiller, and W. F. Northrop, "Ammonia-air burner scaling law for industrial scale-up." Manuscript in preparation.

RELEVANT COURSEWORK AND TRAININGS

Heat Transfer, Multiphase Flows, Fluid Mechanics, Computational Fluid Dynamics, Molecular gas Dynamics, Cleanroom Handling, Manufacturing Processes, Mathematical Modeling of Applied Mathematics.